Lecture 1 (September 1st)

Definition. (Tensor product) Let V and W be vector spaces over a common field. A tensor product of V and W is a pair

$$(V \otimes W, \otimes)$$

where $V \otimes W$ is a vector space over the aforementioned field and $\otimes : V \times W \to V \otimes W$ is a bilinear map (called the canonical tensor product map) such that for any linear map $f : V \times W \to Z$, there exists a unique linear map

$$\tilde{f} : V \otimes W \to Z$$

such that

$$f(v, w) = \tilde{f}(v \otimes w) = \tilde{f}(\otimes(v, w))$$

for all $v \in V$ and $w \in W$.

Definition. (Direct sum) Let V and W be vector spaces over a common field. The direct sum of V and W is defined as the vector space

$$V \oplus W := \{(v, w) \,|\, v \in V, w \in W\}$$

over the aforementioned common field.

Definition. (Fock space) Let $\mathcal{H}_1$ be a complex Hilbert space (A complete inner product space) that describes a single particle. The Fock space over $\mathcal{H}_1$, denoted $\mathcal{F}(\mathcal{H}_1)$, is defined as:

$$\mathcal{F}(\mathcal{H}_1) = \bigoplus_{n=0}^{\infty} (\mathcal{H}_1^{\otimes n})_{S,A} = \boldsymbol{C} \oplus \mathcal{H}_1 \oplus (\mathcal{H}_1 \otimes \mathcal{H}_1)_{S,A} \oplus \dots$$

Remark. There are some issues with combining special relativity (SR) and quantum mechanics (QM) naively:

(i) Special relativity treates t and $\mathbf{x}$ as equal compotents of a 4-vector while quantum mechanics quantises $\mathbf{x}$ but not t

(ii) Special relativity allows for creation and annhilation of particles whereas quantum mechanics doesn't accomodate such processes

The resultion is quantum field theory (QFT) which is done by quantising classical field theory.

Definition. (Special relativity) We deal with the Minkowski space denoted as $\boldsymbol{R}^{1,3}$, characterised by the metric

$$d(x, y) = \eta_{\mu\nu} x^\mu y^\nu$$

where $x^\mu = (ct, x, y, z)$ and $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$. We define automorphisms called the

Lorentz transformations (LT) as

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

where $\Lambda^\mu{}_\rho \eta_{\mu\nu} \Lambda^\nu{}_\sigma = \eta_{\rho\sigma}$. These transformations are those such that leave metric invariant, as seen below

$$ds'^2 = \eta_{\mu\nu} dx'^\mu dx'^\nu = \eta_{\mu\nu} \Lambda^\mu{}_\rho dx^\rho \Lambda^\nu{}_\rho dx^\rho = ds^2$$

In this course, we only deal with proper ($\det \Lambda = 1$) and orthochronous transformations $\Lambda^0{}_0 \geq 1$ (note that $|\Lambda^0{}_0| \geq 1$). In the language of matrices, we denote the (μ, ν) component of η^{-1} as $\eta^{\mu\nu}$ and of Λ^T as $(\Lambda^T)^\nu_\mu$. Using this, we can express the previous identity as

$$\eta = \Lambda^T \eta \Lambda$$

Meanwhile, Lorentz tensors (scalars, contravariant, and covariant) are defined as objects that transform accordingly. A general rank (p, g) tensor is then naturally defined as an object that transforms as

$$T'^{\mu'_1 \dots \mu'_p}{}_{\nu'_1 \dots \nu'_q}$$

Note that we can also define ∂_μ and ∂^μ that transform covariantly and contravariantly respectively. You'll need to check that η and its inverse lowers and raises indexes respectively. Depending on their norms, we can define vectors to be either time-like, space-like, and null/light-like.

Definition. (Representation of the Lorentz group) Are there representation of fields beyond tensors that transform a specific way under Lorentz transformations? Yes! Spinors! Consider the Lie algebra of the Lie group of Lorentz tranformations, where

$$\Lambda^\mu{}_\nu = \exp(i\omega^\mu{}_\nu) = \delta^\mu{}_\nu + i\omega^\mu{}_\nu + \mathcal{O}(\omega^2)$$

Using the definition of Lorentz transformations, we come to the conclusion that the algebra should satisfy

$$\omega_{\mu\nu} + \omega_{\nu\mu} = 0$$

There are 6 linearly independent 4×4 anti-symmetric matrices that satisfy such equation and we want to express them as a combination of transformation parameters and 6 independent Lorentz generators in vector representation.

$$\omega^\mu{}_\nu = \frac{1}{2} \alpha_{\rho\sigma} (M^{\rho\sigma})^\mu{}_\nu$$

where of course $M^{\rho\sigma} = -M^{\sigma\rho}$. With this freedom, we have

$$(M^{01})_{\mu\nu} = \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

In general,

$$(M^{\rho\sigma})^{\mu}{}_{\nu} = -i(\eta^{\rho\mu}\delta^{\rho}{}_{\nu} - \eta^{\rho\mu}\delta^{\rho}{}_{\nu})$$

The commutation relations of Lorentz generators in vector representations thus become

$$[M^{\mu\nu}, M^{\rho\sigma}] = i(\eta^{\mu\rho}M^{\nu\sigma} + \eta^{\nu\rho}M^{\mu\rho} - \eta^{\mu\rho}M^{\nu\rho} - \eta^{\nu\rho}M^{\mu\rho})$$

Which is called the Lorentz algebra. The Lorentz algebra is more fundamental than the specific form of generators in the specific representation. General Lorentz transformations in temrs of Lorentz generators thus become

$$\Lambda = \exp\left(\frac{i}{2}\alpha_{\mu\nu}M^{\mu\nu}\right)$$

$\alpha_{\mu\nu} \in \boldsymbol{R}$ $M^{\mu\nu} = (M^{\mu\nu})^{\dagger}$ thus $\Lambda^{\dagger} = \Lambda$. A natural question is – what do $M^{\mu\nu}$ look like in the spinor representation?

Lecture 2 (September 3rd)

Recall. We have learned the Lorentz algebra that satisfies the commutator relations

$$[M^{\mu\nu}, M^{\rho\sigma}] = i\Big(\eta^{\mu\rho}M^{\nu\sigma} + \eta^{\nu\sigma}M^{\mu\rho} - \eta^{\mu\sigma}M^{\nu\rho} - \eta^{\nu\rho}M^{\mu\sigma}\Big)$$

where

$$\Lambda = \exp\left[\frac{i}{2}\alpha_{\mu\nu}M^{\mu\nu}\right]$$

Proposition. (Irreducible representation of the Lorentz algebra) We now want seek to find a specific representation of the Lorentz algebra, suitable for distinguishing how different particles are acted by them. First, we find the individual components of the matrix to be denoted by

$$J^i = \frac{1}{2}\epsilon^{ijk}M^{jk} \qquad K^i = M^{0i}$$

The Lie algebra becomes

$$[J^i, J^j] = i\epsilon^{ijk}J^k \qquad [J^i, K^j] = i\epsilon^{ijk}K^k \qquad [K^i, K^j] = -i\epsilon^{ijk}J^k$$

From here, we can complexify the Lorentz algebra and express

$$J^{\pm i} = \frac{1}{2}(J^i \pm iK^i)$$

What about the commutator relations?

$$[J^{\pm i}, J^{\pm j}] = i\epsilon^{ijk}J^k \qquad [J^{\pm i}, J^{\mp j}] = 0$$

Note that through this complexification process, we are essentially expressing the Lorentz algebra, the tangent space of the Lorentz group at e, through the real $\mathfrak{su}(2)$ component and an imaginary $\mathfrak{su}(2)$ component given the homomorphism

$$\mathfrak{so}(1,3) \simeq \mathfrak{su}(2) \oplus \mathfrak{su}(2)$$

With all the information above, we can simply write the general Lorentz transformation through the parameters

$$\vec{\alpha} = (\alpha_{23}, \alpha_{31}, \alpha_{12}) \quad \vec{\beta} = (\alpha_{01}, \alpha_{02}, \alpha_{03})$$

where

$$\Lambda = \exp\left[\frac{i}{2}\alpha_{\mu\nu}M^{\mu\nu}\right] = \exp\left[i\vec{\alpha}\cdot\vec{J} + i\vec{\beta}\cdot\vec{K}\right] = \exp\left[i(\vec{\alpha} - i\vec{\beta})\cdot\vec{J}^{+}\right]\exp\left[i(\vec{\alpha} + i\vec{\beta})\cdot\vec{J}^{-}\right]$$

Definition. ((j_+, j_-)-representation of the Lorentz transform) Note that by definition, a representation of a Lorentz algebra is a Lie algebra homomorphism

$$\phi : \mathfrak{so}(3,1)_{\boldsymbol{C}} \to \mathrm{End}(V)$$

where V is the representation space. Note that a Lie algebra homomorphism is a map between two Lie algebras that respects both the vector space structure and the bracket.

$$\phi([X,Y]) = [\phi(X), \phi(Y)]$$

We now create these target vector spaces using ladder operators. We now define a (j_+, j_-) representation space of the Lorentz transformations to be the vector space with the basis

$$\{m_+, m_- \mid m_\pm = -j_\pm, -j_\pm + 1, \ldots, j_\pm\}$$

(here, $j_\pm \in \boldsymbol{Z}_{\geq 0}/2$) constructed through the ladder operators

$$J^+_\pm = J^{+1} \pm iJ^{+2} \qquad J^-_\pm = J^{-1} \pm iJ^{-2}$$

satisfying

$$J^+_\pm|m_+, m_-\rangle \sim |m_+ \pm 1, m_-\rangle \qquad J^-_\pm|m_+, m_-\rangle \sim |m_+, m_- \pm 1\rangle$$

We would also have

$$\sum_{i=1}^{3}(J^\pm)^2|m_+, m_-\rangle = j_\pm(j_\pm + 1)|m_+, m_-\rangle$$

Example. (1D (0,0)-representation as spin-0 scalars) We see that for this representation, all Lorentz transforms are of the form

$$\Lambda|_{(0,0)} = \mathbf{1}$$

and that the vector space elements $|0,0\rangle = \phi(x)$ transform as

$$\phi'(x') = \Lambda|_{(0,0)}\phi(x) = \phi(x)$$

Example. (2D (1/2,0)-representation as spin-1/2 left-chiral spinors) We see that for this representation, all Lorentz transforms are of the form

$$\Lambda|_{(1/2,0)} = \exp\left[i(\vec{\alpha} - i\vec{\beta} \cdot \vec{J}^{+})\right]$$

As we are dealing with a 2D complex vector space, an arbitrary transformation is a linear combination of the Pauli matrices, and in matrix notation,

$$l_{+} = \exp\left[i(\vec{\alpha} - i\vec{\beta}) \cdot \frac{\vec{\sigma}}{2}\right]$$

and vector space elements $|\pm j, 0\rangle = \psi_{+}(x)$ transform as

$$\psi'_{+}(x') = \Lambda|_{(1/2,0)}\psi_{+}(x) = l_{+}\psi_{+}(x)$$

Example. (2D (0,1/2)-representation as spin-1/2 right-chiral spinors) We see that for this representation, all Lorentz transforms are of the form

$$\Lambda|_{(0,1/2)} = \exp\left[i(\vec{\alpha} - i\vec{\beta} \cdot \vec{J}^{-})\right]$$

As we are dealing with a 2D complex vector space, an arbitrary transformation is a linear combination of the Pauli matrices, and in matrix notation,

$$l_{-} = \exp\left[i(\vec{\alpha} - i\vec{\beta}) \cdot \frac{\vec{\sigma}}{2}\right]$$

and vector space elements $|0, \pm J\rangle = \psi_{-}(x)$ transform as

$$\psi'_{-}(x') = \Lambda|_{(0,1/2)}\psi_{-}(x) = l_{-}\psi_{-}(x)$$

Definition. (Chirality and helicity) We call a vector left-chiral or right-chiral based on which $SU(2)$ factor of the Lorentz group it transforms by. Helicity is a property of fields.

Lemma. We find that

(i) Choosing how the left-chiral spinors transform automatically dictate how the right-chiral spinors transform, as
$$(\vec{J}^{\pm})^{\dagger} = \vec{J}^{\mp}$$

(ii) The map between left and right-chiral spinors manifest through the fact that

$$(l_\pm)^\dagger = (l_\mp)^{-1} \qquad \& \qquad (l_\pm)^* = \sigma_2 l_\mp \sigma_2$$

as $SU(2)$ elements. We see that

$$\sigma_2 \psi'_\pm(x')^* = l_\mp \sigma_2 \psi_\pm(x)^*$$

Therefore, conjugation is the exact map between left and right-chiral spinors.

Definition. (Invariant tensors) We define

$$\begin{cases} \sigma^\mu = (I_2, \vec{\sigma}) \\ \bar{\sigma}^\mu = (I_2, -\vec{\sigma}) \end{cases}$$

which leads

$$\begin{cases} (l_-)^+ \sigma^\mu l_- = \Lambda^\mu{}_\nu \sigma^\nu \\ (l_+)^+ \bar{\sigma}^\mu l_+ = \Lambda^\mu{}_\nu \bar{\sigma}^\nu \end{cases}$$

Therefore the Lorentz generators for the $\{(1/2, 0), (0, 1/2)\}$ representations in terms of σ^μ and $\bar{\sigma}^\mu$ can be expressed as

$$\begin{cases} (\frac{1}{2}, 0) : M^{0i} = -\frac{i}{2}\sigma^i, M^{ij} = \frac{1}{2}\varepsilon^{ijk}\sigma^k \end{cases}$$

Lecture 3 (September 8th)

Definition. (Grassmann numbers) Grassman numbers are numbers, contrarily to complex numbers, that describe fermions. Grassmann variables are tuplets

$$\{\theta_1, \ldots, \theta_n\}$$

that satisfy

$$\{\theta_i, \theta_j\} = \theta_i\theta_j + \theta_j\theta_i = 0$$

The Grassmann algebra is defined as

$$\Lambda(v) = \boldsymbol{C} \oplus V \oplus V^2 \oplus \ldots \oplus V^n$$

where V is an n-dimensional vector space with the basis $\{\theta_i\}$ over the field of complex numbers. Basic operations are defined as

(i) Addition $\theta_i + \theta_j$

(ii) Complex multiplication $c\theta_i$ $(c \in \boldsymbol{C})$

(iii) Product $\theta_i\theta_j$

The general element of the Grassmann algebra is called the Grassmann number,

$$\begin{aligned} f(\theta) =& f_0 + \sum_{i_1=1}^{n} f_{i_1}\theta_{i_1} + \frac{1}{2}\sum_{i_1,i_2=1}^{n} f_{i_1 i_2}\theta_{i_1}\theta_{i_2} + \dots \\ =& \sum_{k=0}^{n}\frac{1}{k!}\sum_{i_1,\dots,i_k=1}^{n} f_{i_1\dots i_k}\theta_{i_1}\dots\theta_{i_k} \end{aligned}$$

Definition. (Differentiation of Grassmann variables)

$$\frac{\partial\theta_j}{\partial\theta_i} = \delta_{ij} \qquad \frac{\partial\mathbf{1}}{\partial\theta_i} = 0 \qquad \frac{\partial}{\partial\theta_i}(\theta_j\theta_k) = \delta_{ij}\theta_k$$

Definition. (Integration of Grassmann variables)

$$\int d\theta\, f(\theta) = \frac{\partial}{\partial\theta} f(\theta)$$

Notice that once we change the integration measure from $\theta'_i = A_{ij}\theta_j$,

$$\begin{aligned} \int d\theta_1 \dots d\theta_n\, f(\theta) =& \frac{\partial}{\partial\theta_1}\cdots\frac{\partial}{\partial\theta_n} f(\theta) \\ =& A_{i_1 1}\dots A_{i_n n}\frac{\partial}{\partial\theta'_{i_1}}\cdots\frac{\partial}{\partial\theta'_{i_n}} f(A^{-1}\theta') \\ =& \varepsilon_{i_1\dots i_n} A_{i_1 1}\dots A_{i_n n}\frac{\partial}{\partial\theta'_1}\cdots\frac{\partial}{\partial\theta'_n} f(A^{-1}\theta') \\ =& \det(A)\int d\theta'_1\dots d\theta'_n\, f(A^{-1}\theta') \end{aligned}$$

which gives us

$$d\theta'_1\dots d\theta'_n = (\det A)^{-1} d\theta_1\dots d\theta_n$$

as

$$\frac{\partial}{\partial\theta_i} = \frac{\partial\theta'_j}{\partial\theta_i}\frac{\partial}{\partial\theta'_j}$$

Definition. (Gaussian integral wth respect to two sets of Grassman variables)

$$\begin{aligned}
&\int d\bar{\theta}_1 d\theta_1 \dots d\bar{\theta}_n d\theta_n \, e^{-\sum_{i,j=1}^n \bar{\theta}_i M_{ij} \theta_j} \\
&= \int d\bar{\theta}_1 d\theta_1' \dots d\bar{\theta}_n d\theta_n' (\det M) e^{o \sum_{i=1}^n \bar{\theta}_i \theta_i'} \\
&= \det M \prod_{i=1}^n \int d\bar{\theta}_i d\theta_i' \, e^{-\bar{\theta}_i \theta_i'} \\
&= \det M \prod_{i=1}^n \int d\bar{\theta}_i d\theta_i' (1 - \bar{\theta}_i \theta_i') = \det M
\end{aligned}$$

(where $M_{ij}\theta_j = \theta_i'$)

Definition. (Functional derivatives) Let X be a space of admissible functions $y : [a, b] \to \boldsymbol{R}$ like the Banach space $L^2([a, b])$ or $C^1([a, b])$. A functional is a mapping $F : X \to \boldsymbol{R}$ defined as

$$\frac{\delta F[f]}{\delta f(x_0)} = \lim_{\varepsilon \to 0} \frac{F[f(x) + \varepsilon \delta(x - x_0)] - F[f(x)]}{\varepsilon}$$

Example. (Examples of functional derivatives) We have

(i) $\int f^n$ gives

$$\frac{\delta F[f]}{\delta f(z)} = n(f(z))^{n-1}$$

(ii) $\int (f')^n$ gives

$$\frac{\delta F[f]}{\delta f(z)} = -nf^{(n)}(z)$$

(iii) $\int dx \, K(y, x) f(x)$ gives

$$\frac{\delta F[f]}{\delta f(z)} = K(y, z)$$

(iv) $f(y)$ gives

$$\frac{\delta F[f]}{\delta f(z)} = \delta(y - z)$$

Classical Field Theory

Theorem. (Lagrangian formalism) Classically, the action S of a system is written as

$$S[x, \dot{x}] = \int_{t_i}^{t_f} dt \, L(x_i, \dot{x}_i)$$

For a field, we write

$$S[\phi, \dot{\phi}] = \int_{t_i}^{t_f} dt\, L[\phi, \dot{\phi}]$$

where $\phi = \phi(t, \vec{x}) = \phi(x)$. The variation of this action is given as

$$\delta S[\phi, \dot{\phi}] = \int dt \int d^3\vec{x} \Big[\frac{\delta L}{\delta \phi(x)} \delta\phi(x) + \frac{\delta L}{\delta \dot{\phi}(x)} \delta\dot{\phi}(x)\Big]$$

Through the application of Hamilton's principle (the variation of the action is zero for an arbitrary variation of the field such that $\delta\phi(t_i, \vec{x}) = \delta\phi(t_f, \vec{x}) = 0$) we obtain

$$\begin{aligned} 0 &= \int d^4x \Big(\frac{\delta L}{\delta\phi(x)} \delta\phi(x) + \frac{\delta L}{\delta\dot{\phi}(x)} \frac{\partial}{\partial t} \delta\phi(x)\Big) \\ &= \int d^4x \Big(\frac{\delta L}{\delta\phi(x)} - \frac{\partial}{\partial t} \frac{\delta L}{\delta\dot{\phi}(x)}\Big) \delta\phi(x) + \int d^4x\, \frac{\partial}{\partial t}\Big(\frac{\delta L}{\delta\dot{\phi}(x)} \delta\phi(x)\Big) \end{aligned}$$

where the second boundary terms vanish to zero. The Lagrangian in "local" relativistic classical field theory leads to the concept of a Lagrangian density.

$$S[\phi, \dot{\phi}] = \int dt\, L[\phi, \dot{\phi}] = \int d^4x\, \mathcal{L}(\phi(x), \dot{\phi}(x), \nabla\phi(x)) = \int d^4x\, \mathcal{L}(\phi(x), \partial_\mu\phi(x))$$

Alternative expressions in local classical field theory include

$$\frac{\delta L}{\delta \phi} = \frac{\partial \mathcal{L}}{\partial \phi} - \nabla\Big(\frac{\partial \mathcal{L}}{\partial(\nabla\phi)}\Big) \qquad \frac{\delta L}{\delta\dot{\phi}} = \frac{\partial \mathcal{L}}{\partial\dot{\phi}}$$

and the Euler-Lagrangian equations as

$$0 = \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\Big)$$

as

$$\begin{aligned} 0 &= \int d^4x \Big[\frac{\partial \mathcal{L}}{\partial\phi} d\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \partial_\mu(\delta\phi)\Big] \\ &= \int d^4x \Big(\frac{\partial \mathcal{L}}{\partial\phi} - \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\Big)\Big)\delta\phi + \int d^4x\, \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi\Big)\Big) \end{aligned}$$

assuming that $\phi, \partial_\mu\phi \to 0$ fast enough as $|\vec{x}| \to \infty$.

Theorem. (Hamiltonian formulation) The canonically conjugate field is defined as

$$\pi(x) = \frac{\delta L}{\delta\dot{\phi}(x)} = \frac{\partial \mathcal{L}}{\partial\dot{\phi}(x)}$$

The Hamiltonian is then defined as

$$H[\phi, \pi] = \int d^3\vec{x}\, \Big(\pi(x)\dot{\phi}(x) - \mathcal{L}\Big) = \int d^3x\, \mathcal{H}$$

The Eular-Lagrange equations then become Hamilton's equations of motion

$$\dot{\phi}(x) = \frac{\delta H}{\delta \pi(x)} \qquad \dot{\pi}(x) = -\frac{\delta H}{\delta \phi(x)}$$

Lecture 4 (September 10th)

Definition. (Poisson bracket) The Poisson bracket for two functions $F = F[\phi, \pi]$ and $G = G[\phi, \pi]$ is defined as

$$\{F, G\}_{PG} = \int d^3\vec{x} \Big(\frac{\delta F}{\delta \phi(x)} \frac{\delta G}{\delta \pi(x)} - \frac{\delta F}{\delta \pi(x)} \frac{\delta G}{\delta \phi(x)} \Big)$$

Notice that using the Poisson bracket, Hamilton's equation of motion becomes

$$\begin{cases} \dfrac{d\phi}{dt} = \dfrac{\delta H}{\delta \pi} = \{\phi, H\}_{PB} \\[2ex] \dfrac{d\pi}{dt} = -\dfrac{\delta H}{\delta \phi} = \{\phi, H\}_{PB} \end{cases}$$

Then, for an arbitrary functional F,

$$\begin{aligned} \frac{dF}{dt} &= \int d^3\vec{x} \Big(\frac{\delta F}{\delta \phi(x)} \dot{\phi}(x) + \frac{\delta F}{\delta \pi(x)} \dot{\pi}(x) \Big) \\ &= \int d^3\vec{x} \Big(\frac{\delta F}{\delta \phi} \frac{\delta H}{\delta \pi} + \frac{\delta F}{\delta \pi} \Big(- \frac{\delta H}{\delta \phi} \Big) \Big) = \{F, G\}_{PB} \end{aligned}$$

We have some special cases,

$$\begin{aligned} \{\phi(t, \vec{x}), \phi(t, \vec{y})\}_{PB} =& 0 = \{\pi(t, \vec{x}), \pi(t, \vec{y})\}_{PB} \\ \{\phi(t, \vec{x}), \pi(t, \vec{y})\}_{PB} =& \int d^3 z \, \frac{\delta \phi(t, \vec{x})}{\delta \phi(t, \vec{z})} \frac{\delta \pi(t, \vec{y})}{\delta \pi(t, \vec{z})} = \int d^3 z \, \delta^{(3)}(\vec{x} - \vec{z}) \delta^{(3)}(\vec{y} - \vec{z}) = \delta^{(3)}(\vec{x} - \vec{y}) \end{aligned}$$

Definition. (Spin-0 Klein-Gordan Fields) The Klein-Gordan (KG) action is given as

$$S_{KG} = \int d^4 x \, \underbrace{\Big(- \frac{1}{2} \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} m^2 \phi^2 \Big)}_{\mathcal{L}_{KG}}$$

The Klein-Gordan equation that comes from this action is

$$0 = \frac{\partial \mathcal{L}_{KG}}{\partial \phi} - \partial_\mu \Big(\frac{\partial \mathcal{L}_{KG}}{\partial(\partial_\mu \phi)} \Big) = -m^2 \phi - \partial_\mu (-\partial^\mu \phi) = (\partial^2 - m^2) \phi$$

here, $\partial^2 = \partial^\mu \partial_\mu = \Box$. What are the solutions to the Klein-Gordan equation? The simplest case is given as a plane wave solution

$$\phi_{\pm,\vec{p}}(x) = A_\pm e^{i(\vec{p}\cdot\vec{x}\pm E_p t)}$$

where $E_p = \sqrt{\vec{p}^2 + m^2}$. Is it genuinely a solution? Notice that

$$(\partial^2 - m^2)\phi_{\pm,\vec{p}}(x) = (-\partial_t^2 + \nabla^2 - m^2)\phi_{\pm,\vec{p}} = (E_p^2 - \vec{p}^2 - m^2)\phi_{\pm,\vec{p}} = 0$$

The general soltuion is then given by a Fourier expansion,

$$\phi(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}}\Big(A_-(\vec{p})e^{i(\vec{p}\cdot\vec{x}-E_p t)} + A_+(-\vec{p})e^{-i(\vec{p}\cdot\vec{x}-E_p t)}\Big)$$

where we will now write $\vec{p}\cdot\vec{x} - E_p t = p_\mu x^\mu$. Imposing the condition that $\phi(x) = \phi^*(x)$,

$$\phi(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}}\Big(a(\vec{p})e^{ip_\mu x^\mu} + a^*(\vec{p})e^{-p_\mu x^\mu}\Big)$$

The canonical conjugate field is then automatically given as

$$\begin{aligned}\pi(x) =& \frac{\partial \mathcal{L}_{KG}}{\partial\dot{\phi}(x)} = \dot{\phi}(x)\\ =& \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}}(-iE_p)(a^*(\vec{p})e^{ip\cdot x} - a(\vec{p})^* e^{-ip\cdot x}\end{aligned}$$

The Hamiltonian density becomes

$$\mathcal{H}_{KG} = \pi\dot{\phi} - \mathcal{L}_{KG} = \frac{1}{2}(\pi^2 + (\nabla\phi)^2 + m^2\phi^2)$$

The Poisson brackets become

$$\begin{cases}\{\phi(t,\vec{x}),\phi(t,\vec{y})\}_{PB} = \{\pi(t,\vec{x}),\pi(t,\vec{y})\}_{PB} = 0\\ \{\phi(t,\vec{x}),\pi(t,\vec{y})\}_{PB} = \delta^{(3)}(\vec{x}-\vec{y})\end{cases}$$

Definition. (Spin-1/2 Dirac fields) There is no first order, Lorentz covariant equation of motion for spin-0 fields, but we can find one for spin-1/2 fields. We begin from the Weyl equations of motion:

$$\begin{cases}i\bar{\sigma}^\mu\partial_\mu\psi_+ + m\sigma_2\psi_+^* = 0\\ i\sigma^\mu\partial_\mu\psi_- + m\sigma_2\psi_-^* = 0\end{cases}$$

Notice that without the absence of math, they require both left and right chirals. A problem with the equations is that there is no $U(1)$ symmetry.

$$\psi_\pm \to \psi_\pm' = e^{i\theta_\pm}\psi_\pm$$

Coupling the two Weyl equations of motion with doubled degrees of freedom, we have

$$\begin{cases} 0 = i\bar{\sigma}^\mu \partial_\mu \psi_+ + m\psi_- \\ 0 = i\sigma^\mu \partial_\mu \psi_- + m\psi_+ \end{cases} \iff \left(\begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \partial_\mu - m \right) \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = 0$$

The Dirac action that yields this equation is

$$S_D = \int d^4x\, \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$$

where $\gamma^\mu \partial_\mu = \not{\partial}$. The dirac equation is therefore

$$0 = \frac{\partial \mathcal{L}_D}{\partial \bar{\psi}} - \partial_\mu \Big(\frac{\partial \mathcal{L}_D}{\partial(\partial_\mu \bar{\psi})} \Big) = (i\not{\partial} - m)\psi$$

A natural equation would be to ask what would happen for $\bar{\psi}$. The elementary solution as a plane wave is given as

$$\psi_{\pm,\vec{p}}(x) = u_\pm(\vec{p}) e^{\pm i p_\mu x^\mu}$$

where $u_\pm(\vec{p})$ is expected to be a 4×1 matrix with $(\not{p} \pm m)u_\pm(\vec{p}) = 0$. We can check that for our elementary solution.

$$(i\not{\partial} - m)\psi_{\pm,\vec{p}}(\vec{x}) = (i(\pm i\gamma^\mu p_\mu) - m)\psi_{\pm,\vec{p}} = \mp(\not{p} \pm m)\psi_{\pm,\vec{p}} = 0$$

Note that also, $p^\mu p_\mu + m^2 = 0$.

$$\begin{aligned} 0 =& -(\gamma^\nu p_\nu \mp m)(\gamma^\mu p_\nu \pm m)u_\pm(\vec{p}) = -(\gamma^\nu \gamma^\mu p_\nu p_\mu - m^2)u_\pm(\vec{p}) \\ =& -\Big(\frac{1}{2}\{\gamma^\nu, \gamma^\mu\} p_\nu p_\mu - m^2 \Big) u_\pm(\vec{p}) = -(-p^\mu p_\mu - m^2)u_\pm(\vec{p}) = 0 \end{aligned}$$

where the commutators equal $-2\eta^{\mu\nu}$. Using the Dirac basis we can express the four independent solutions for $u_\pm$ (2 for each sign) as

$$0 = (\not{p} \pm m)u_\pm(\vec{p}) \propto \begin{pmatrix} \pm m - E_p & \vec{p}\cdot\vec{\sigma} \\ -\vec{p}\cdot\vec{\sigma} & \pm m + E_p \end{pmatrix} \begin{pmatrix} \phi_\pm \\ \chi_\pm \end{pmatrix}$$

This implies that

$$\begin{cases} \chi_+ = & \dfrac{\vec{p}\cdot\vec{\sigma}}{m + E_p}\phi_+ \\ \phi_- = & \dfrac{\vec{p}\cdot\vec{\sigma}}{m + E_p}\chi_- \end{cases}$$

The two independent u_+ become (for $i = 1, 2$)

$$u^i_+(\vec{p}) = \sqrt{m + E_p} \begin{pmatrix} \phi^i_+ \\ \dfrac{\vec{p}\cdot\vec{\sigma}}{m + E_p}\phi^i_+ \end{pmatrix}$$

and two independent for u_- become

$$u^i_-(\vec{p}) = \sqrt{m + E_p} \begin{pmatrix} \dfrac{\vec{p}\cdot\vec{\sigma}}{m + E_p}\chi^i_- \\ \chi^i_- \end{pmatrix}$$

with

$$\begin{cases} (\phi^i_+)^\dagger \phi^j_+ = \delta^{ij} \\ \sum_{i=1}^2 \phi^i_+ (\phi^i_+)^\dagger = I_2 \end{cases}$$

and the same for χ^i_-. A good example of vectors that do this are

$$\phi^j_+ = \{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\}$$

Remark. (Properties of $u^i_\pm(\vec{p})$) We have orthonormality

$$\bar{u}^i_\pm(\vec{p}) u^j_\pm(\vec{p}) = \pm 2m\delta^{ij}$$

and

$$\bar{u}^i_\pm(\vec{p}) u^j_\mp(\vec{p}) = 0$$

and the spin state sum

$$\sum_{i=1}^2 u^i_\pm \bar{u}^i_\pm(\vec{p}) = -(\not{p} \mp m)$$

Other useful properties include

$$\begin{cases} u^i_\pm{}^\dagger(\vec{p}) u^j_\pm(\vec{p}) = 2E_p\delta^{ij} \\ u^i) \pm (\vec{p})^\dagger u^j_\mp(-\vec{p}) = 0 \end{cases}$$

The general solution then

Lecture 5 (September 15th)

Definition. (Spin-1 Maxwell fields) The Maxwell action is given as

$$S_M = \int d^4x \, \underbrace{\Big(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \Big)}_{\mathcal{L}_M}$$

Note that for a vector field $A_\mu(x)$, the field strength is defined as

$$F_{\mu\nu} = 2\partial_{[\mu}A_{\nu]} = 2\Big(\frac{\partial_\mu A_\nu - \partial_\nu A_\mu}{2!}\Big)$$

Notice that S_M has a certain gauge symmetry, where the following shift leaves the action invariant:

$$A_\mu \to A_\mu + \partial_\mu \Lambda$$

There are typical choices for Λ. Two are the Coulomb ($\nabla \cdot \vec{A} = 0$) and the Lorentz gauge ($\partial_\mu A^\mu = 0$). From these, we get the Maxwell's equations (equations of motion)

$$\begin{aligned}
0 =& \frac{\partial \mathcal{L}_M}{\partial A_\mu} - \partial_\nu\Big(\frac{\partial \mathcal{L}_{\mathcal{M}}}{\partial(\partial_\nu A_\mu)}\Big) \\
=& -\partial_\nu\Big(-\frac{1}{2}F^{\rho\sigma}\frac{\partial F_{\rho\sigma}}{\partial(\partial_\nu A_\mu)}\Big) \\
=& -\partial_\nu\Big(-\frac{1}{2}F^{\rho\sigma}(\delta^\nu_\rho\delta^\mu_\sigma - \delta^\mu_\rho\delta^\nu_\sigma)\Big) \\
=& -\partial_\nu(-F^{\nu\mu})
\end{aligned}$$

From the gauge symmetry, we have the Bianchi identity, $\partial_{[\mu}F_{\nu\rho]} = 0$ which implies the no-magnetic monopole and Faraday's law, and the equation of motion yield the Gauss and Ampere law.

Proposition. We provide the general solution through the plane wave solution, which is obtainable under the Lorentz gauge $\partial_\mu A^\mu = 0$.

$$A^r_{\mu,\vec{p}}(x) = \epsilon^r_\mu(\vec{p})e^{ip\cdot x}$$

where $r \in \{1, 2\}$ with $p_\mu p^\mu = p^\mu \epsilon^r_\mu(\vec{p}) = 0$. Observe that

$$\partial^\mu A^r_\mu(x) = ip^\mu\epsilon^r_\mu(\vec{p})e^{ip\cdot x} = 0$$

From this, we can look at the deriviative of the field strength to conclude that our identity indeed holds.

$$\partial^\mu F^r_{\mu\nu} = \partial^\mu\partial_\mu A^r_\nu - \partial^\mu\partial_\nu A^r_\mu = -p^\mu p_\mu \epsilon^r_\mu(\vec{p})e^{ip\cdot x} = 0$$

Definition. (Polarisation vectors) There are two physical ones, where the Lorentz gauge kills one degree of freedom.

$$\varepsilon^r_\mu(\vec{p}) = (\varepsilon^r_0, \vec{\varepsilon}^r) = \Big(-\frac{\vec{p}\cdot\vec{\varepsilon}^r}{p^0}, \vec{\varepsilon}^r\Big)$$

where we have used $p^\mu\varepsilon^r_\mu(\vec{p}) = p^0\varepsilon^r_0 + \vec{p}\cdot\vec{\varepsilon}^r = 0$. There is also the residual gauge from the fact that $\varepsilon^r_\mu(\vec{p}) \to \varepsilon^r_\mu(\vec{p}) + \alpha p_\mu$ which kills the other.

$$\varepsilon^r_\mu(\vec{p}) = \Big(-\frac{\vec{p}\cdot\vec{\varepsilon}^r}{p^0} - \alpha p^0, \vec{\varepsilon}^r + \alpha\vec{p}\Big) = \Big(0, \vec{\varepsilon}^r - \frac{\vec{p}\cdot\vec{\varepsilon}^r}{(p^0)^2}\vec{p}\Big)$$

where $\alpha = -\vec{p}\cdot\vec{\varepsilon}^r/(p^0)^2$. Which implies $(0, \vec{\varepsilon}^{1,2}(\vec{p}))$. We often choose the orthonormal ones,

$$\vec{\varepsilon}^r(\vec{p}) \cdot (\vec{\varepsilon}^s(\vec{p}))^* = \delta^{rs}$$

There are two unphysical ones,

$$\begin{cases} \varepsilon^0_\mu(\vec{p}) = n_\mu = (1,0,0,0) \\ \varepsilon^3_\mu(\vec{p}) = (0,\hat{p}) = \dfrac{p_\mu + n_\mu(p\cdot n)}{|p_\mu + n_\mu(p\cdot n)|} \end{cases}$$

These are necessary to describe virtual photons and are useful in analysing quantum electrodynamic scattering processes.

Theorem. The relations between polarisation vectors are as follows

$$\eta^{\mu\nu}(\varepsilon^r_\mu(\vec{p}))^*\varepsilon^s_\nu(\vec{p}) = \eta^{rs}$$

orthonormality and

$$\sum_{r=0}^{3} \eta_{rr}\varepsilon^r_\mu(\vec{p})\varepsilon^s_\nu(\vec{p})^* = \eta_{\mu\nu}$$

completeness. The above gives

$$\sum_{r=1}^{2} \eta_{rr}\varepsilon^r_\mu(\vec{p})(\varepsilon^s_\nu(\vec{p}))^* = \eta_{\mu\nu} - \frac{p_\mu p_\nu + (p_\mu n_\nu + p_\nu n_\mu)(p\cdot n)}{(p\cdot n)^2}$$

Proposition. Finally, we have

$$A_\mu(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E_p}} \sum_{r=1}^{2} \Big(a^r(\vec{p})\varepsilon^r_\mu(\vec{p})e^{ip\cdot x} + a^r(\vec{p})^*\varepsilon^r_\mu(\vec{p})^* e^{-p\cdot x}\Big)$$

and the canonically conjugate fields and Hamiltonian density is

$$\pi_\mu = \frac{\partial\mathcal{L}_\mathcal{M}}{\partial(\partial_0 A^\mu)} = F_{0\mu} \qquad \& \qquad \mathcal{H}_M = \pi^\mu\partial_0 A_\mu - \mathcal{L}_M$$

Theorem. (Noether's theorem) Conservation laws in classical field theory are given by Noether's theorem. Consider an infinitesimal transformation,

$$\phi(x) \to \phi'(x) = \phi(x) = \delta\phi(x)$$

where δ is called a continuous symmetry of the action

$$S[\phi] = \int d^4x\, \mathcal{L}(\phi(x), \partial_\mu \phi(x))$$

The consequence of this is that there is suppose to be a conserved (Noether) current. Directly evaluating $\delta\mathcal{L}$,

$$\begin{aligned}\delta\mathcal{L} =& \frac{\partial \mathcal{L}}{\partial \phi}\,\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\,\partial_\mu(\delta\phi) \\ =& \Big(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\Big)\,\delta\phi + \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\,\delta\phi\Big)\end{aligned}$$

Invariance of the action for a continuous symmetry ($\delta S = 0$) is dictated by the condition

$$\delta\mathcal{L} = \partial_\mu f^\mu$$

where $f^\mu \to 0$ fast enough at infinity. Identifying the above, we have, by defining

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\,\delta\phi - f^\mu$$

the following

$$\partial_\mu j^\mu = 0$$

on-shell. The Noether current above is conserved on-shell! What is the conserved quantity?

$$Q = \int d^3\vec{x}\, j^0$$

as

$$\frac{dQ}{dt} = \int d^3x\, \partial_0 j^0 = \int d^3\vec{x}\, \nabla \cdot \vec{j} = -\int d\vec{a} \cdot \vec{j} = 0$$

There are mutliple types of symmetries. (1) Spacetime symmetries are symmetries that involve transformations of spacetime coordinates. There are also (2) internal symmetries, which are symmetries that act on fields but do not affect spacetime symmetries themselves. For these symmetries, we have $U(1)$, $SU(N)$ symmetries and etcetera.